

Product Reference Manual (V1.0)

Description

The module integrates the MP34DT05-A digital MEMS microphone in a compact development format designed for fast prototyping and embedded audio applications. It features a standard PDM microphone configuration with direct access to the CLK, DAT, SEL, 3V, and GND signals for simple integration with microcontrollers and audio processing systems.

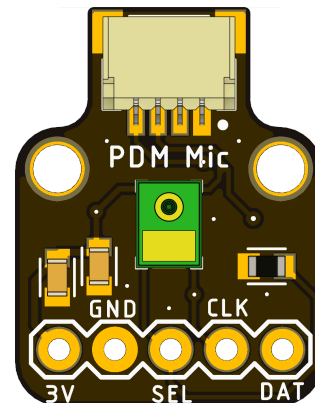
The board includes a JST 1.0 mm connector for compact cable-based connections and an additional 2.54 mm header interface for direct protoboard or jumper-wire integration. This dual-connector approach allows flexible use in portable devices, evaluation systems, educational projects, and rapid hardware validation environments.

The module is intended for low-power digital audio acquisition applications such as voice detection, sound monitoring, speech recognition, and IoT development, while maintaining a compact footprint suitable for quick experimental setups and embedded designs.

DevLab format compatibility.

Simplicity and compatibility are the primary objectives of the DevLab form factor. It offers a board layout that is compact and optimized for serial communication, which includes I²C and SPI interfaces.

This format enables the establishment of rapid and dependable connections via a Protocol I²C connector that is entirely compliant with the Qwiic and STEMMA standards.



Key Features

- Digital MEMS microphone
- MP34DT05-A audio sensor
- PDM digital audio output
- Omnidirectional audio sensing
- Low power consumption
- Signal-to-noise ratio: 64 dB
- Sensitivity: -26 dBFS ±3 dB
- Acoustic overload point: 122.5 dB SPL
- Supply voltage: 1.6 V to 3.6 V
- EMI shielded package
- Compact DevLab form factor
- JST 1.0 mm connector
- 2.54 mm prototyping header interface
- Breadboard compatible
- Rapid embedded integration

Hardware Features

- MP34DT05-A MEMS microphone
- PDM clock input interface
- PDM digital data output
- Left/Right channel selection pin
- 3.3 V compatible operation
- Compact PCB layout
- Low-profile top-port microphone package
- Integrated power filtering capacitors

- Through-hole and JST connectivity
- Embedded audio acquisition support

Software Support

The module is primarily intended for development platforms with native or flexible PDM audio acquisition support.

Recommended platforms include:

- Raspberry Pi Pico SDK
- Arduino for RP2040
- Arduino for RP2350
- Pico SDK for RP2040/RP2350
- PlatformIO
- Zephyr RTOS

Recommended compatible devices:

- RP2040
- RP2350
- ARM Cortex-M platforms with native PDM peripherals
- FPGA and programmable logic systems

The RP2040 and RP2350 platforms are especially recommended due to their flexible PIO architecture, allowing reliable high-speed PDM signal acquisition and software-based audio processing implementations.

Direct support on standard ESP32 platforms is limited because most ESP32

Variants do not provide native standalone PDM microphone acquisition interfaces compatible with this module without additional configuration layers or external audio processing methods.

Applications

- Embedded systems
- Voice recognition
- Portable media devices
- IoT audio systems
- Smart speakers
- Gaming and VR systems
- Digital cameras
- Educational electronics project

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1 The Board

1.1 Accessories

- 1×5-pin 2.54 mm male header
- 1x QWIIC 4-pin JST 1.0 mm Harness

2 Ratings

Symbol	Description	Min	Typ	Max	Unit
VDD	Supply voltage	1.6	3.3	3.6	V
IDD	Current consumption	-	650	-	μA
IDDPDN	Power-down current	-	5	-	μA
FCLK	PDM clock frequency	1.2	2.4	3.25	MHz
TOP	Operating temperature	-40	25	85	°C
SNR	Signal-to-noise ratio	-	64	-	dB
AOP	Acoustic overload point	-	122.5	-	dB SPL
SO	Sensitivity	-29	-26	-23	dBFS
VIH	Logic high input voltage	$0.65 \times VDD$	-	$VDD + 0.3$	V
VIL	Logic low input voltage	-0.3	-	$0.35 \times VDD$	V
CLOAD	Output capacitive load	-	-	100	pF
TON	Startup time	-	10	-	ms
ESD	Electrostatic protection (HBM)	-	±2000	-	V

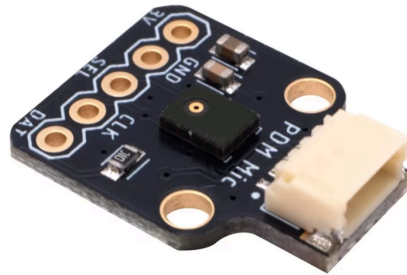
2.1 Recommended Operating Conditions

Note:

Typical values are based on MP34DT05-A standard operating conditions at 25 °C with a 2.4 MHz PDM clock.

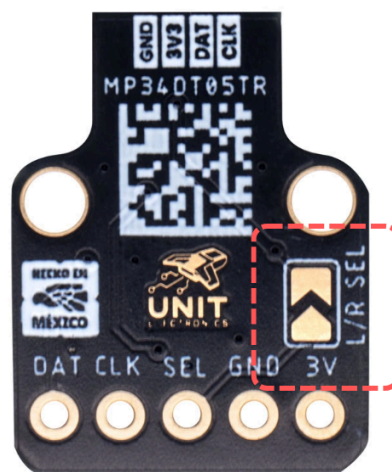
3 Functional Overview

The module is designed as a compact plug-and-play digital microphone solution for rapid embedded audio integration. It incorporates the MP34DT05-A MEMS microphone and provides a JST 1.0 mm connector interface compatible with compact 4-wire cable systems for power and digital signal transport.



The board operates using a standard PDM (Pulse Density Modulation) audio interface composed of a clock input signal (CLK) and a digital audio output signal (DAT). The host system generates the clock signal while the microphone transmits synchronized digital audio data through the DAT line.

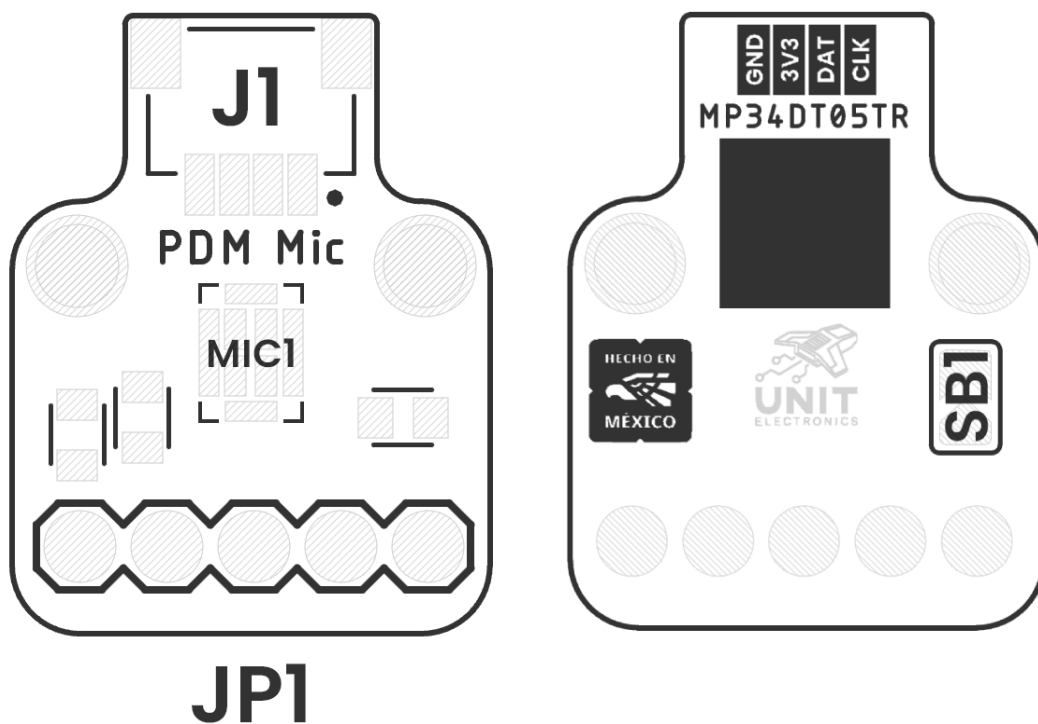
A dedicated SEL pin is included to configure the microphone operating channel. This pin allows selection between the left or right PDM channel configuration, enabling mono or stereo microphone implementations when multiple modules are used together.



The module is optimized for simplified wiring and fast implementation in embedded systems, educational platforms, audio acquisition systems, and rapid prototyping environments. The JST connector format reduces wiring complexity while improving mechanical reliability during development and experimental integration.

The CLK signal defines the synchronization frequency required for microphone operation and audio sampling. The DAT signal carries the PDM audio stream that can later be converted into PCM audio data through software filtering or DSP processing inside the host microcontroller.

3.2 Board Topology



Top Side

Bottom Side

Views of Board Topology

Views of Topology

Table 3.2.1 - Components Overview

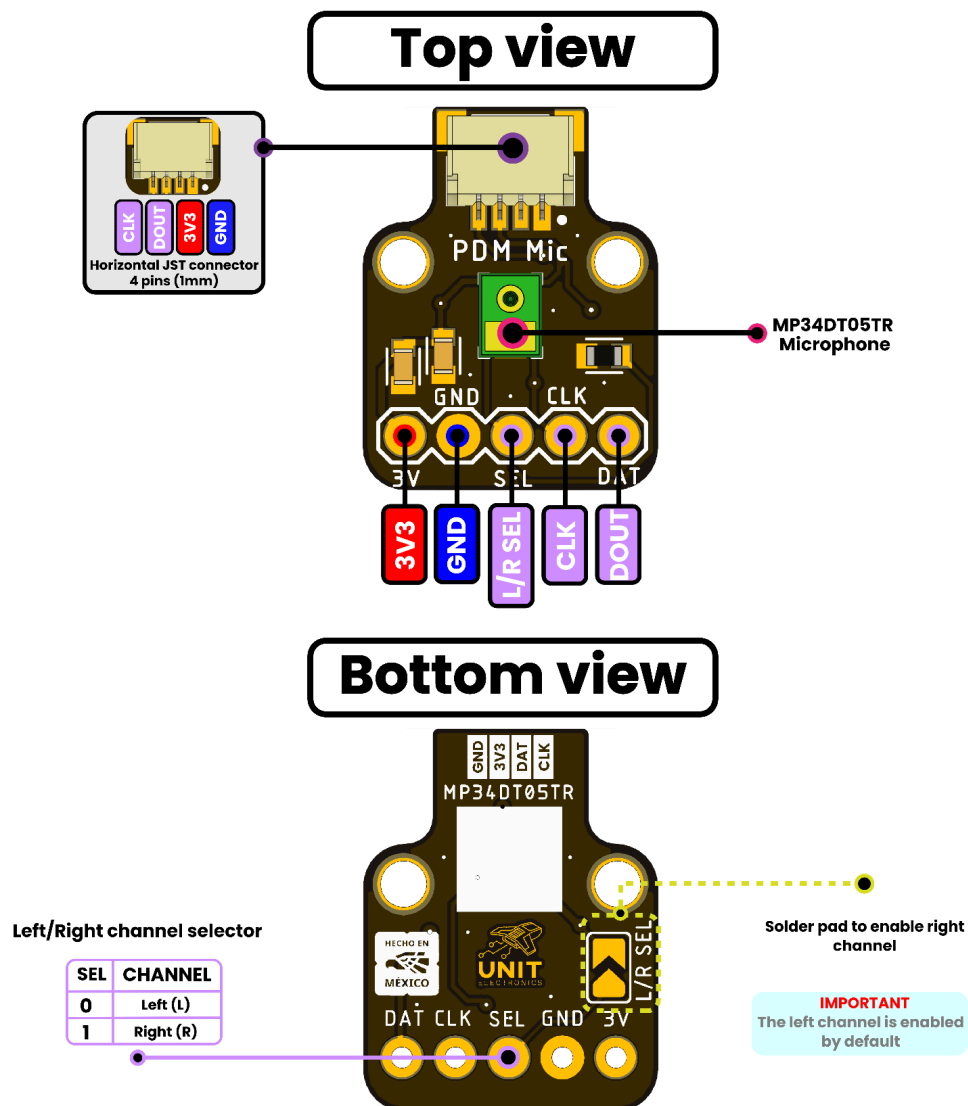
Ref.	Description
MIC1	MP34DT05-A digital MEMS microphone
SB1	SEL solder bridge for Left/Right PDM channel selection
J1	JST 1.0 mm connector for compact cable connection
JP1	2.54 mm external pin header interface

4 Connectors & Pinouts

4.1 General Pinout

PINOUT

DevLab: PDM MP34DT05TR Microphone



DevLab MP34DT05 Sensor Module General Pinout

4.2 Pinout General Description

Pin	Label	Description	Typical Use
1	3V3	Main power supply input for the microphone module	3.3 V regulated power
2	GND	Common ground reference	System ground connection
3	SEL	Left/Right PDM channel selection input	Stereo microphone configuration
4	CLK	PDM synchronization clock input generated by the host	Audio sampling clock
5	DOUT	Digital PDM audio data output	Audio acquisition interface

Left/Right Channel Selection

The SEL pin determines the active PDM channel configuration used by the microphone:

SEL State	Channel
0	Left (L)
1	Right (R)

The module is configured for the left channel by default. A solder bridge located on the bottom side of the board can be modified to enable the right channel configuration for stereo audio applications.

JST Connector Interface

The module includes a horizontal JST 1.0 mm 4-pin connector designed for compact cable interconnection in embedded systems and portable devices.

The JST connector transports:

- CLK signal
- DOUT signal
- 3V3 power
- GND reference

This connector format simplifies rapid integration while reducing wiring complexity and improving mechanical reliability during development and prototyping.

5 Board Operation

The following example demonstrates how to capture audio from the UNIT DevLab MP34DT05TR PDM microphone using RP2040 or RP2350 boards with the Arduino environment.

The implementation is based on the standard `pdm_microphone` example adapted for the DevLab PDM microphone module and uses the PIO-based PDM support included in the arduino-pico core.

The example captures signed 16-bit PCM audio samples and streams them through the Serial interface for visualization and processing.

Wiring Diagram

RP2040 / RP2350 GPIO	PDM Microphone Pin
3.3 V	3V3
GND	GND
GND	SEL
GPIO 4	DOUT
GPIO 5	CLK

Note:

The module is configured for the left audio channel by default when SEL is connected to GND.

How It Works

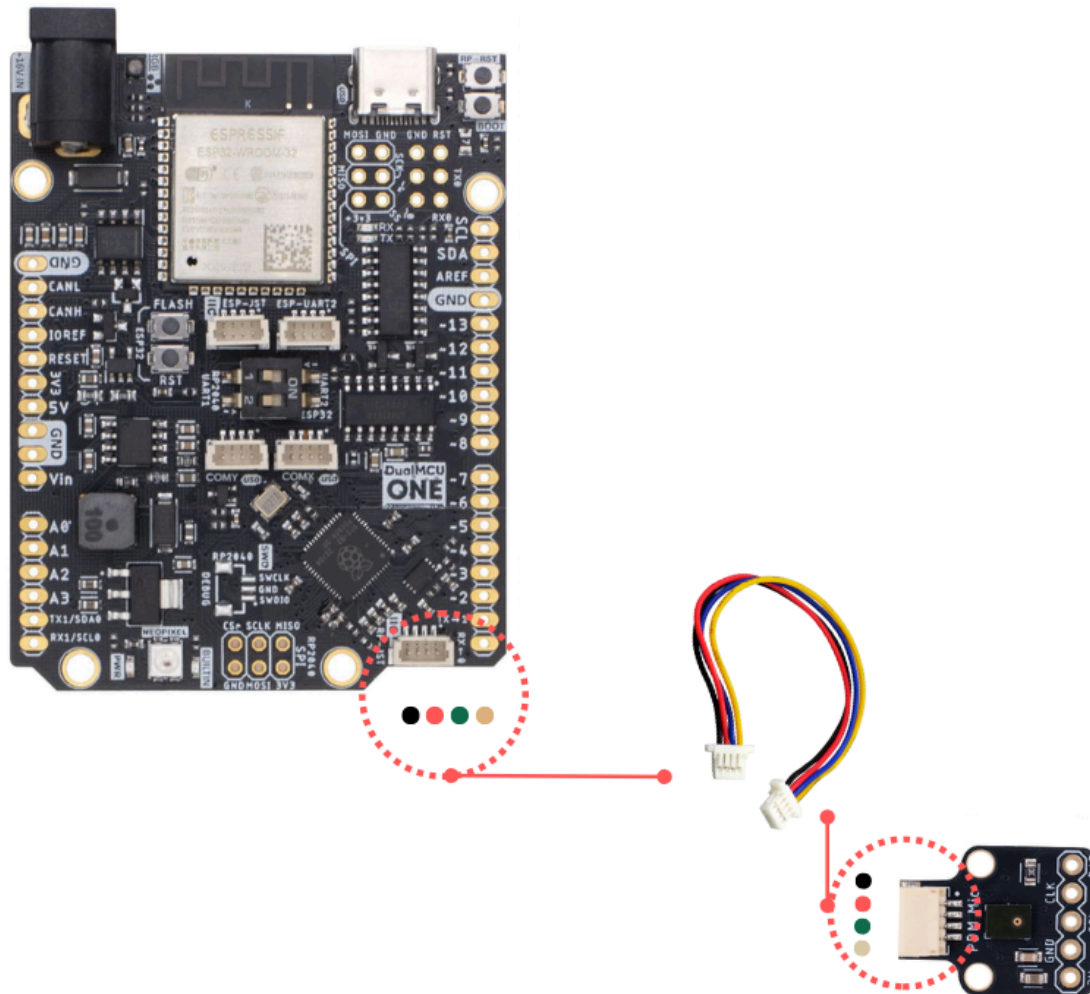
The arduino-pico PDM library configures the RP2040/RP2350 PIO peripheral to receive the PDM audio stream generated by the microphone.

An interrupt callback continuously fills an audio buffer every time new audio data becomes available. The application loop copies the samples and transmits them as signed 16-bit integers through the Serial interface at 115200 baud.

A companion Python visualization script can read the Serial stream and display:

- Real-time waveform
- Amplitude histogram
- Audio statistics
- Signal monitoring

The RP2040 and RP2350 architectures are especially suitable for PDM microphone acquisition due to their flexible PIO subsystem and deterministic timing behavior.



Pin Configuration

The example defines the following pins:

```
const int kPdmDinPin = 4;
const int kPdmClkPin = 5;
```

These values can be modified to match different hardware configurations.

Requirements

Software

- Arduino IDE 2.x
- arduino-pico board package
- Python 3.x (optional for live visualization)

Arduino Board Manager URL

https://raw.githubusercontent.com/UNIT-Electronics/Uelectronics-RP2040-Arduino-Package/main/package_Uelectronics_rp2040_index.json

Supported Boards

- RP2040
- RP2350
- Raspberry Pi Pico
- Raspberry Pi Pico 2
- UNIT DualMCU RP2040
- UNIT RP2350 platforms

Installing the Arduino Core

1. Open Arduino IDE
2. Go to:

File → Preferences

3. Add the Board Manager URL:

https://raw.githubusercontent.com/UNIT-Electronics/Uelectronics-RP2040-Arduino-Package/main/package_Uelectronics_rp2040_index.json

4. Open:

Tools → Board → Boards Manager

5. Search for:

rp2040

6. Install the arduino-pico core
7. Select your board
8. Open the example sketch
9. Upload the firmware

Serial Monitor

Open the Serial Monitor using:

```
115200 baud
```

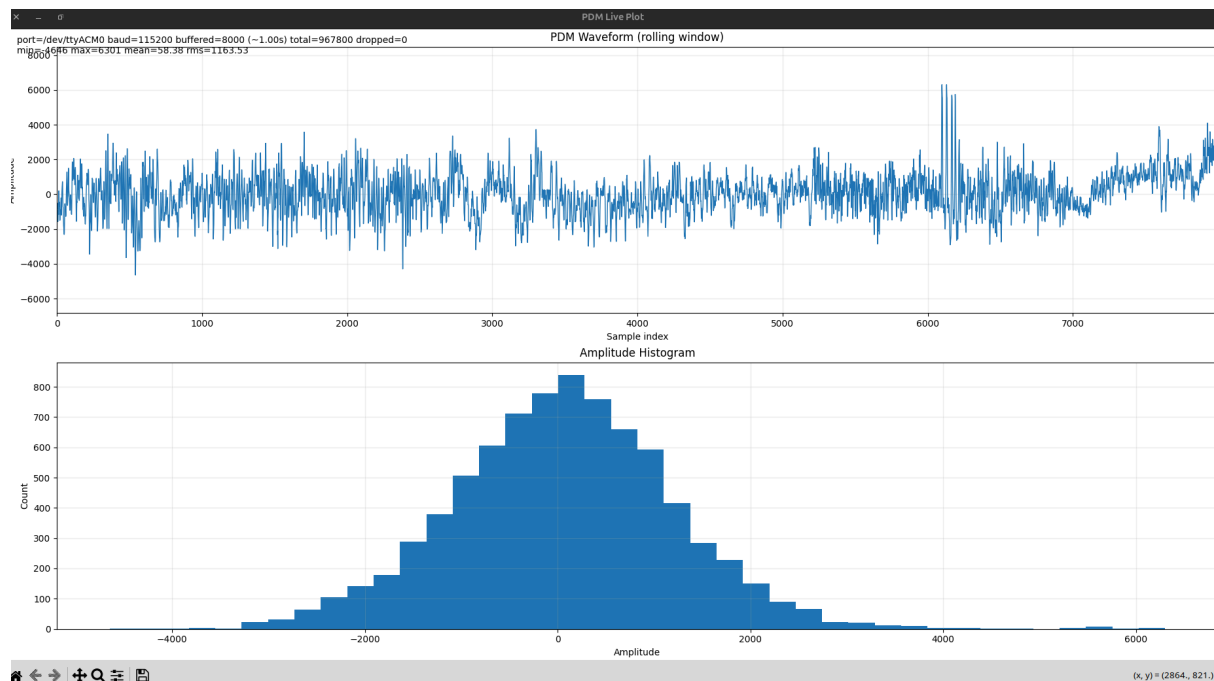
The terminal will display a continuous stream of signed PCM audio samples.

Example:

```
-125
-118
-102
-96
-80
```

Python Live Viewer

The companion Python script allows real-time visualization of the microphone audio stream.



Features

- Rolling waveform plot
- Audio histogram
- RMS calculation
- Signal statistics
- Text-only monitoring mode

Installing Dependencies

```
pip install pyserial matplotlib
```

Running the Viewer

Automatic Port Detection

```
python3 pdm_microphone/serial_live_samples.py
```

Manual Port Selection

```
python3 pdm_microphone/serial_live_samples.py --port /dev/ttyACM0
--baud 115200
```

Extended Rolling Window

```
python3 pdm_microphone/serial_live_samples.py --seconds 2.0
```

Text Mode

```
python3 pdm_microphone/serial_live_samples.py --mode text
```

Script Arguments

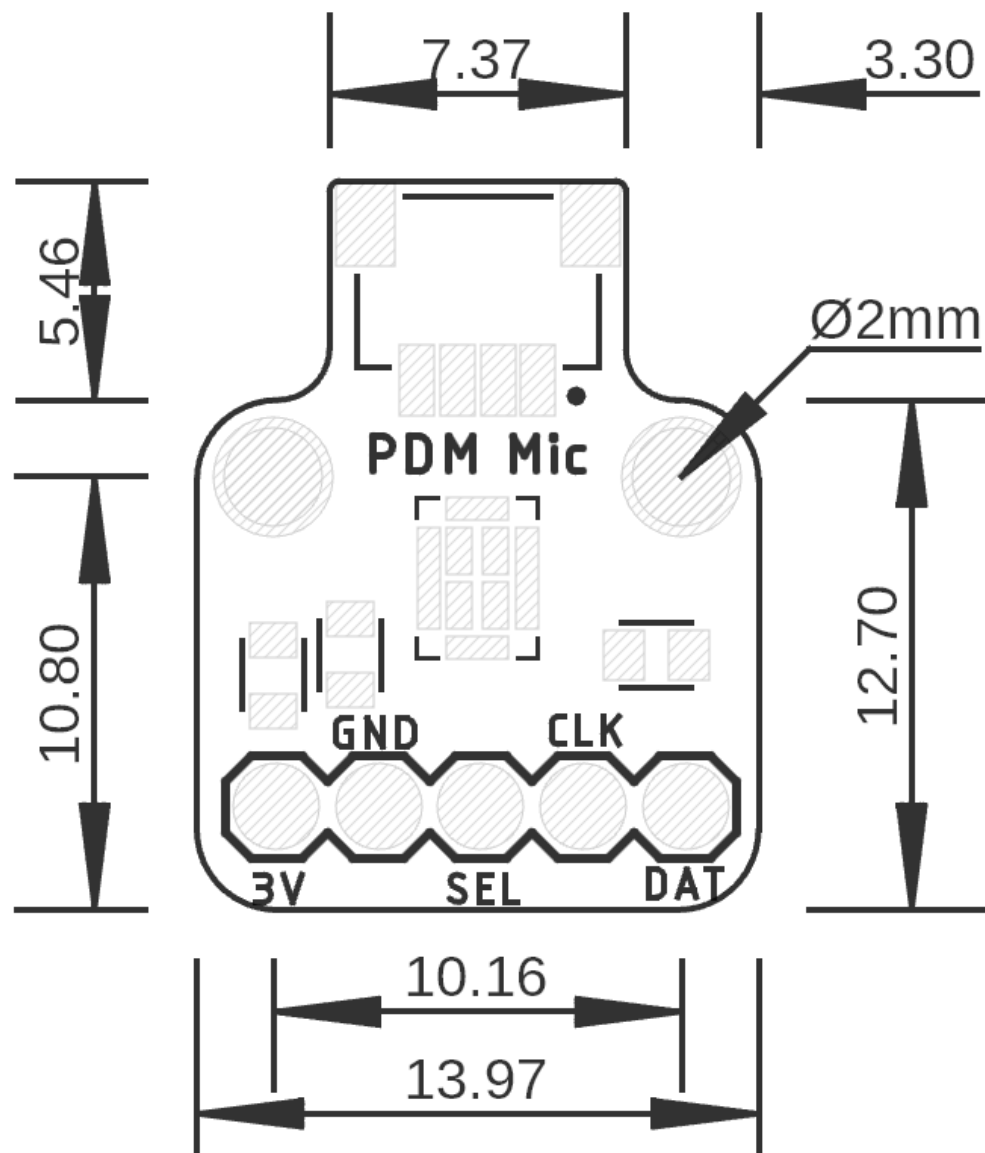
Argument	Default	Description
--port	Auto-detect	Serial device
--baud	115200	Serial baud rate
--sample-rate	8000	Audio sample rate
--seconds	1.0	Rolling window duration
--mode	plot	Plot or text visualization

Notes

- The module uses a standard PDM digital audio interface.
- The RP2040/RP2350 PIO subsystem performs stable high-speed sampling.
- SEL connected to GND enables the left channel.
- Stereo configurations can be implemented using multiple microphones with independent SEL configuration.
- ESP32 platforms are not the primary target for this module due to limited direct standalone PDM acquisition support compared with RP2040/RP2350 platforms.



6. Mechanical Information



Board Dimensions in Millimeters

Mechanical dimensions in millimeters

7 Company Information

Company name	UNIT Electronics
Company website	https://uelectronics.com/
Company Address	Salvador 19, Cuauhtémoc, 06000 Mexico City, CDMX

8 Reference Documentation

Ref	Link
Documentation	https://wiki.uelectronics.com/wiki/unit-mp34dt05tr-a-modulo-microfono-pdm
Github	https://github.com/UNIT-Electronics-MX/unit_devlab_pdm_mp34dt05tr_microphone
Thonny IDE	https://thonny.org/
Arduino IDE	https://www.arduino.cc/en/software
Visual Studio Code	https://code.visualstudio.com/download

9 Appendix

9.1 Schematic

(https://github.com/UNIT-Electronics-MX/unit_devlab_pdm_mp34dt05tr_microphone/blob/main/hardware/unit_sch_v_0_0_1ue0011_devlab_pdm_mp34dt05tr_microphone_en.pdf)

